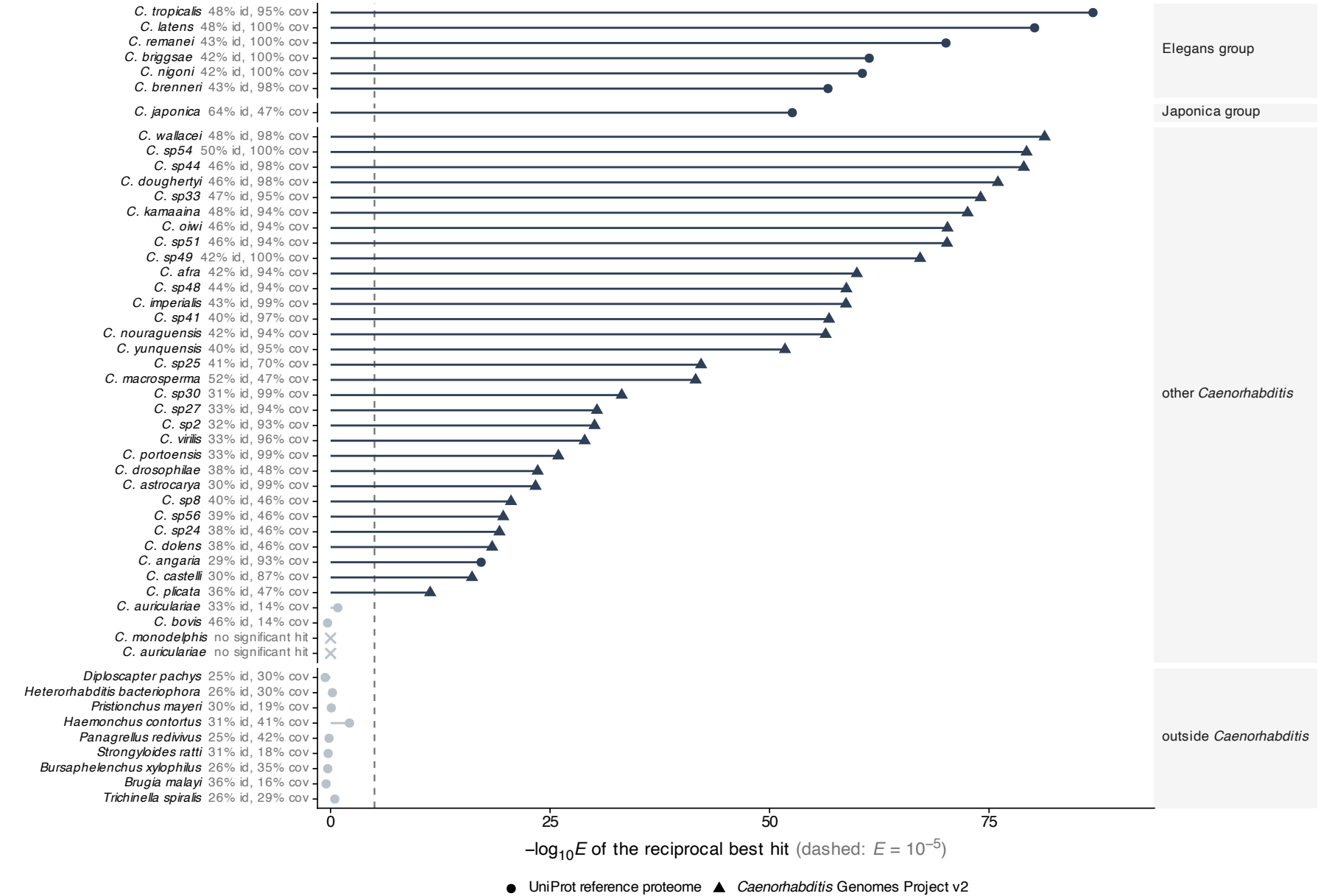


A Outside *Caenorhabditis*, SID-2 has no detectable ortholog

Reciprocal-best-hit blastp of *C. elegans* SID-2 against 52 nematode proteomes, 1,239,391 proteins: 20 UniProt reference proteomes spanning the phylum and 32 from the *Caenorhabditis* Genomes Project v2. The two sets share one species, *C. auriculariae*, so the 52 proteomes are 51 species, and the query is not drawn – leaving 51 comparators, of which 38 clear $E < 1e-5$ and every one is a *Caenorhabditis*. Nothing beyond the genus clears it, including *Diploscapter pachys*, the sister genus, at $E = 4.1$. Four *Caenorhabditis* proteomes also fail, *C. auriculariae* in both of its independent ones; a missing hit in a single proteome is weak evidence about the gene and may only say its annotation is incomplete.



B Constrained positions high, free one lowest

All 173 ectodomain positions (21-193), binned by how many of the 6 full-length Elegans-group orthologs share the *C. elegans* residue. The background is 2.23 of 6 (37%) and only 15% of positions are conserved in all. Calibration that cuts the other way: three of the five ectodomain sequons are intact in all 6 orthologs, so a fully conserved sequon is the norm in this protein and the one at 94 is not a standout.

